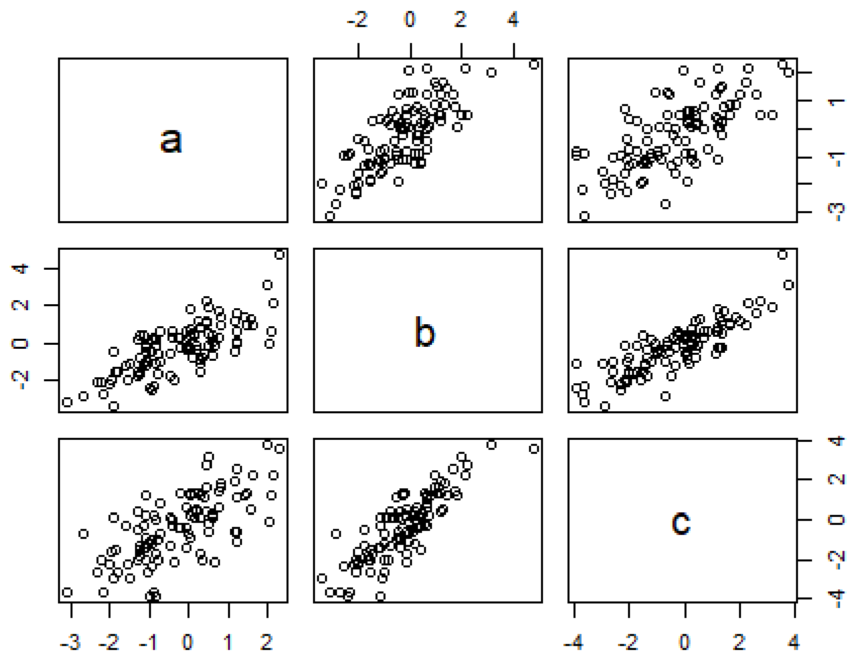
Session1-Exercises.R

```
############################
### Session 1 Exercises ###
############################

#1.1
#a
P <- c(seq(31, 60, 1))
P

##  [1] 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53
54 55
## [26] 56 57 58 59 60

Q <- matrix(seq(31, 60, 1), ncol=5)
Q

##      [,1] [,2] [,3] [,4] [,5]
## [1,]   31   37   43   49   55
## [2,]   32   38   44   50   56
## [3,]   33   39   45   51   57
## [4,]   34   40   46   52   58
## [5,]   35   41   47   53   59
## [6,]   36   42   48   54   60

#b
Qt <- matrix(seq(31, 60, 1), ncol=5, byrow = TRUE)
Qt

##      [,1] [,2] [,3] [,4] [,5]
## [1,]   31   32   33   34   35
## [2,]   36   37   38   39   40
## [3,]   41   42   43   44   45
## [4,]   46   47   48   49   50
## [5,]   51   52   53   54   55
## [6,]   56   57   58   59   60

#1.2
#a
x1 <- rnorm(100)
x2 <- rnorm(100)
x3 <- rnorm(100)

#b
t<- data.frame(a=x1, b=x1+x2, c=x1+x2+x3)
t
```

```
#c
plot(t)
```




```
#sd(t)
#plot created scatter plots pairwise
#sd did not work, columns in a data frame may contain
#different data types, and standard deviation can be
#calculated for numerical data only.

#d
sd(t$a)

## [1] 1.167147

sd(t$b)

## [1] 1.38662

sd(t$c)

## [1] 1.674231

#for everything together:
tmatrix <- as.matrix(t)
sd(tmatrix)

## [1] 1.420492
```

```r
# Attention, it works only when you have the same data type in each column.

#1.3
# Create a vector using a sequence going from 21 to 120. Give it the name
"a".
a<-seq(21,120) # this does the same aa<-21:120

# Define b as the length of a.
b<-length(a)
b

## [1] 100

#Define d the 5th element of the vector a.

d<-a[5]

# Define f the vector containing the elements from the 2nd until the 6th.

f<-a[2:6]

# Define g the vector containing the 1st , 3rd and 7th elements of a.

g<-a[c(1,3,7)]

# Create a vector, call it h, containing the sequence of values of "a" from 1
until 100 every 4 observations.

h<-a[seq(1,100,by=4)]

# Define i the vector containing the elements bigger than 24 and smaller than
29.

i<-a[a>24&a<29]

# Create a matrix (give it the name l) with 25 rows and 4 columns containing
the element of the vector a.

l<-matrix(a,25,4) # this makes the transpose of l:  t(l)

# Define m the vector containing the elements of the second column of l.

m<-l[,2]

# Define n the vector containing the elements of the third row of l.

n<-l[3,]
```

```r
# Define o the vector containing the elements included from row 6 until row
12 and from column 2 until column 3 of l.

o<-l[6:12,2:3]

#1.4
#a
months <- c("Feb", "Jun", "Jul", "Apr", "Aug")
months

## [1] "Feb" "Jun" "Jul" "Apr" "Aug"

#b
sort(months)

## [1] "Apr" "Aug" "Feb" "Jul" "Jun"

## [1] "Apr" "Aug" "Feb" "Jul" "Jun"
#They are in alphabetical order

#c
month_levels <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug",
"Sep", "Oct", "Nov", "Dec")

#d
mymonths <- factor(months, levels = month_levels)
mymonths

## [1] Feb Jun Jul Apr Aug
## Levels: Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec

## [1] Feb Jun Jul Apr Aug
## Levels: Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec

#e
sort(mymonths)

## [1] Feb Apr Jun Jul Aug
## Levels: Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec

## [1] Feb Apr Jun Jul Aug
## Levels: Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
#They are in the order of months, neither in the original order, nor in
alphabetical order.
```